

Project Presentation of Industry-Oriented Hands-On Experience (22CS422)

On

MANNLYTICS: AI-Based Mental Health Detection Using NLP on User-Generated Text



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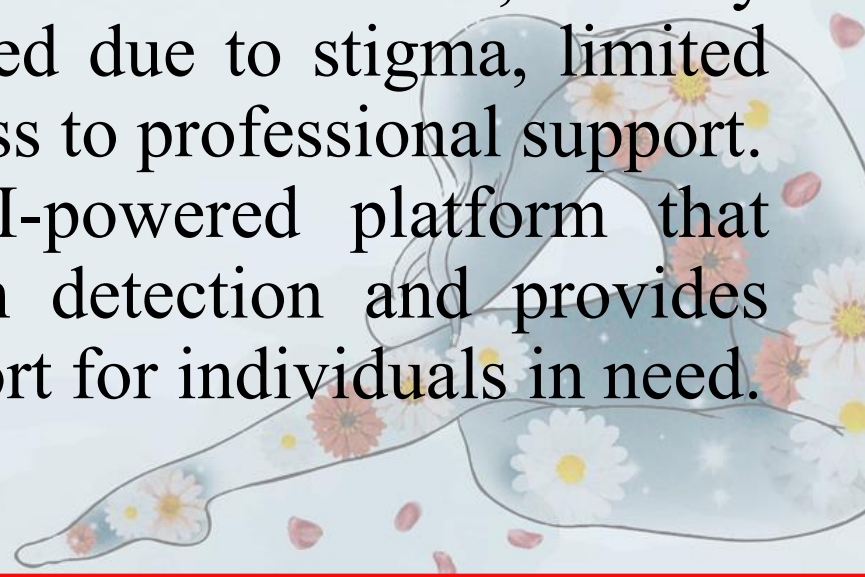
Agenda



- **Introduction**
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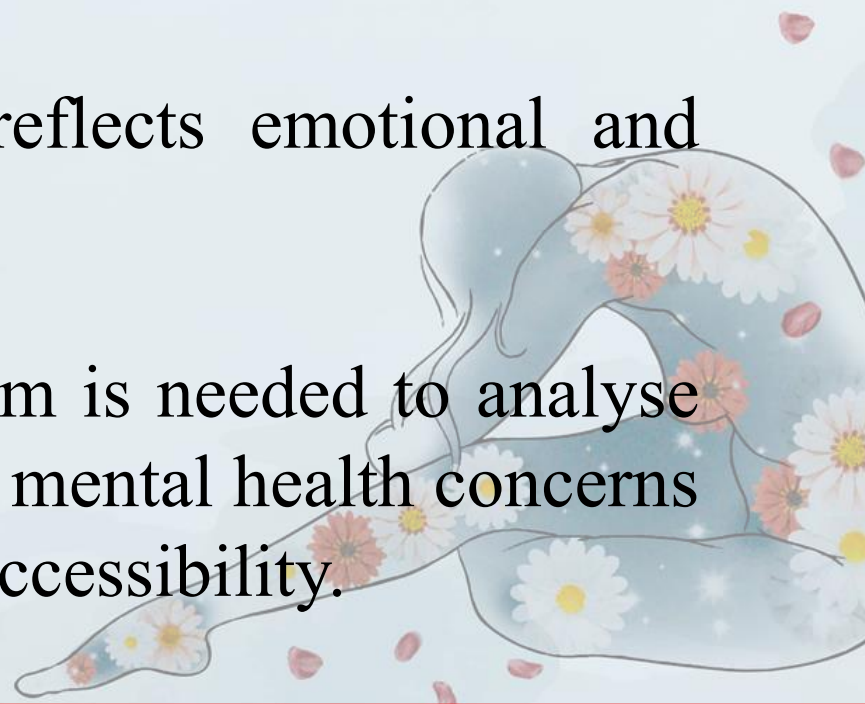


Introduction

- Mental health is a vital component of an individual's emotional, psychological, and social well-being, influencing how people think, feel, and cope with daily challenges.
 - The prevalence of mental health conditions such as anxiety, depression, and stress has increased significantly in recent years. However, many individuals remain undiagnosed due to stigma, limited awareness, and restricted access to professional support.
 - This project presents an AI-powered platform that performs early mental health detection and provides accessible, user-friendly support for individuals in need.
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Problem Statement

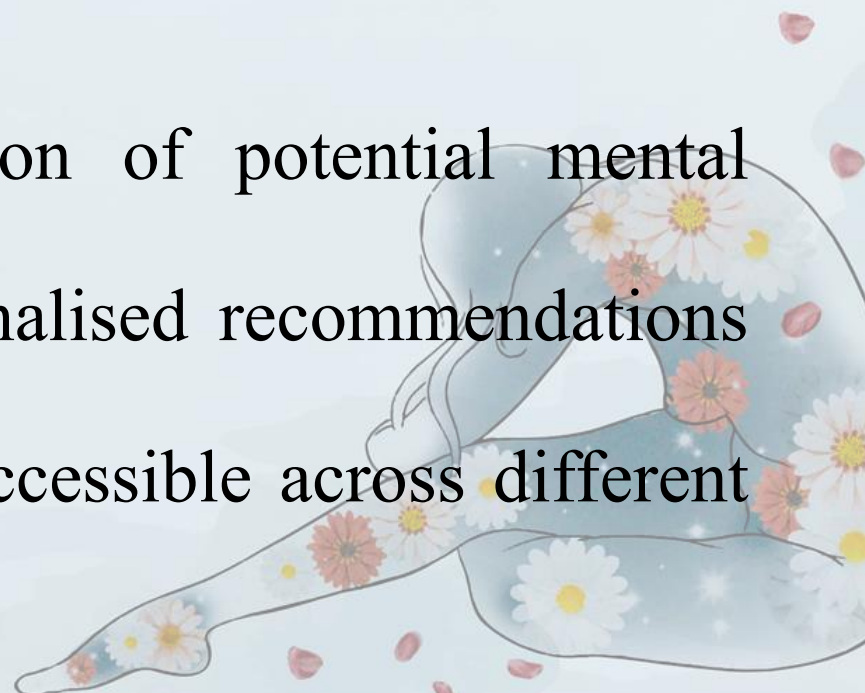
- Mental health disorders affect millions, yet many cases remain undiagnosed.
- Limited access to professionals, stigma, and high diagnostic costs delay timely intervention.
- User-generated text often reflects emotional and psychological patterns.
- An AI and NLP-based system is needed to analyse text and detect early signs of mental health concerns while ensuring privacy and accessibility.



Primary Objectives:

- To design and develop an AI-based mental health detection system
- To create an intuitive and user-friendly web interface
- To ensure the secure handling and storage of user data.

Secondary Objectives:

- To enable early identification of potential mental health issues
 - To provide users with personalised recommendations and guidance
 - To make the system easily accessible across different devices.
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- A decorative illustration in the bottom right corner shows a person lying down, partially covered by a blue cloth, surrounded by various flowers like daisies and tulips. The background is a soft, light blue with some floral patterns.

Proposed Solution

- Web-based platform for early mental health detection using NLP.
- Analyses user-generated text entered through text, voice, or chatbot.
- Uses VADER, TF-IDF, Logistic Regression, DistilBERT, and RoBERTa.
- Predicts mental health categories and calculates confidence and risk scores.
- Displays results on an interactive dashboard with visual insights.
- Includes a Calm Corner module with meditation and stress-relief resources.



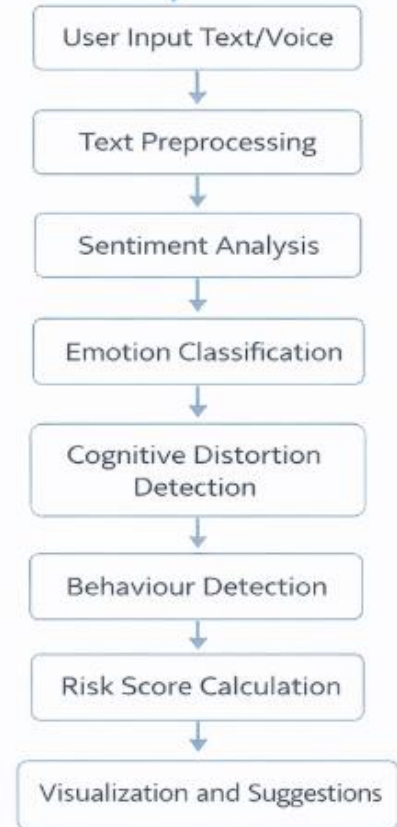
- **Frontend (React.js):** User interface for text input, chatbot, voice input, and dashboard.
 - **Backend (Node.js & Express.js):** API handling and communication between system components.
 - **Machine Learning (Python):** VADER, TF-IDF, Logistic Regression, DistilBERT, and RoBERTa for prediction and risk scoring.
 - **Database (MongoDB):** Secure storage of users, predictions, and resources.
 - **Visualisation & Calm Corner:** Dashboard with charts, recommendations, and relaxation tools.
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Methodology



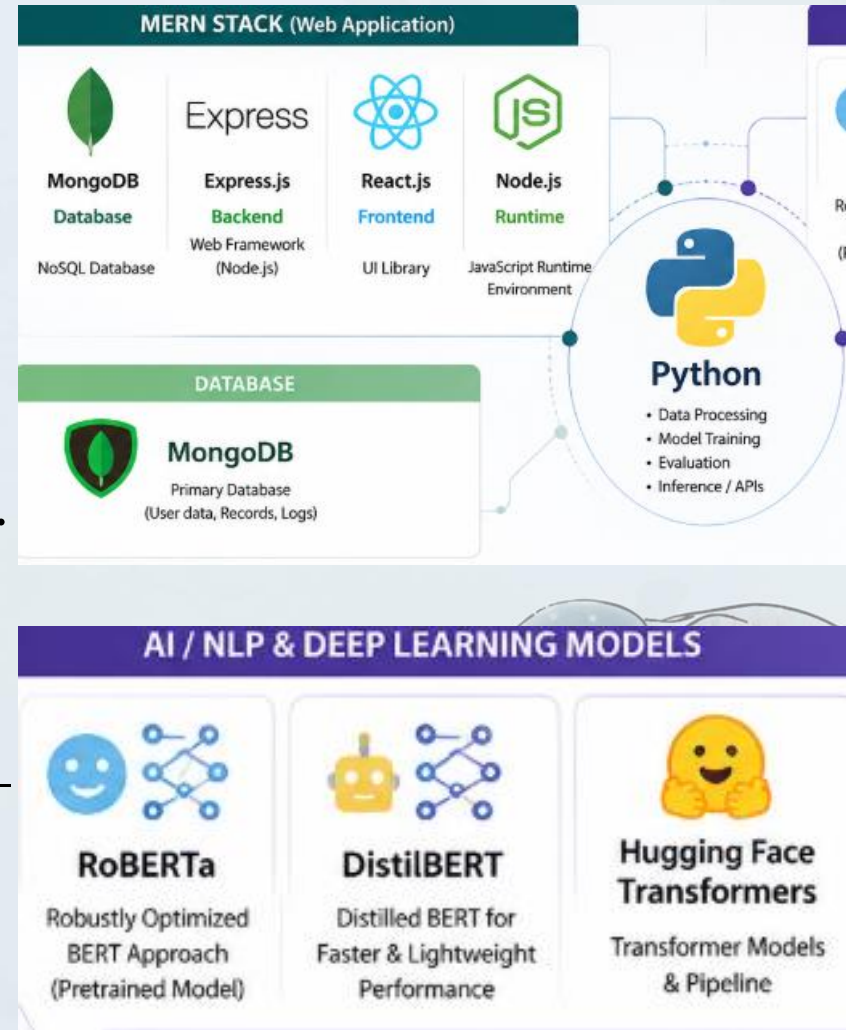
- User registers or logs into the platform.
- User submits text through text input, voice input, or chatbot.
- Backend preprocesses and forwards the input to the NLP model.
- Hybrid NLP models analyse the text and predict the mental health category.
- Confidence and risk scores are generated.
- Results are displayed with recommendations and visual insights.
- Calm Corner resources are provided for additional support.

Mental Health Detection System

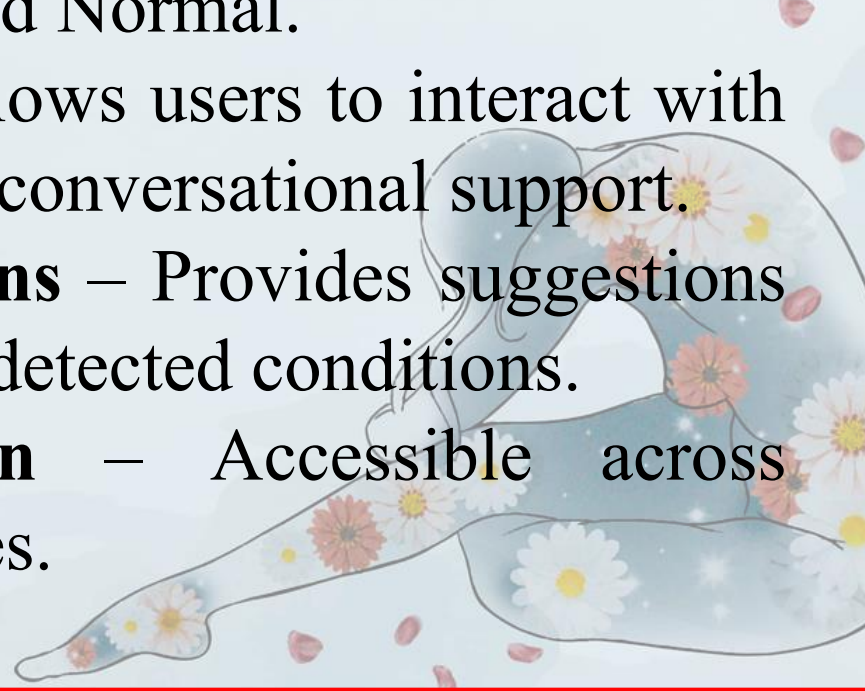


Technology Stack

- **Frontend:** React.js, Tailwind CSS, JavaScript
- **Backend:** Node.js, Express.js.
- **Machine Learning and NLP:**
Python – Used for model development and text processing.
Scikit-learn – Implements TF-IDF and Logistic Regression models.
NLTK / VADER Sentiment Analyzer – Performs sentiment analysis.
Transformers (Hugging Face) – Provides DistilBERT and RoBERTa models for contextual analysis.
- **Database:** MongoDB



Key Features

- **AI-Powered Mental Health Detection** – Analyzes user-generated text using NLP and Machine Learning to identify early signs of mental health concerns.
 - **Multi-Class Classification** – Predicts conditions such as Anxiety, Depression, Stress, Bipolar Disorder, Personality Disorder, Suicidal Tendencies, and Normal.
 - **Chatbot and Voice Input** – Allows users to interact with the platform using text, voice, or conversational support.
 - **Personalised Recommendations** – Provides suggestions and wellness resources based on detected conditions.
 - **Responsive Web Application** – Accessible across desktops, tablets, and smartphones.
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Implementation

- Developed using a modular and component-based architecture.
- Frontend built with React.js and Tailwind CSS.
- Backend implemented using Node.js and Express.js.
- Python-based NLP models integrated for real-time predictions.
- Authentication ensures secure user access.
- MongoDB stores users, predictions, and resources.
- Includes chatbot, voice input, dashboard, and Safe Space features.
- Optimised for performance, responsiveness, and reliability.



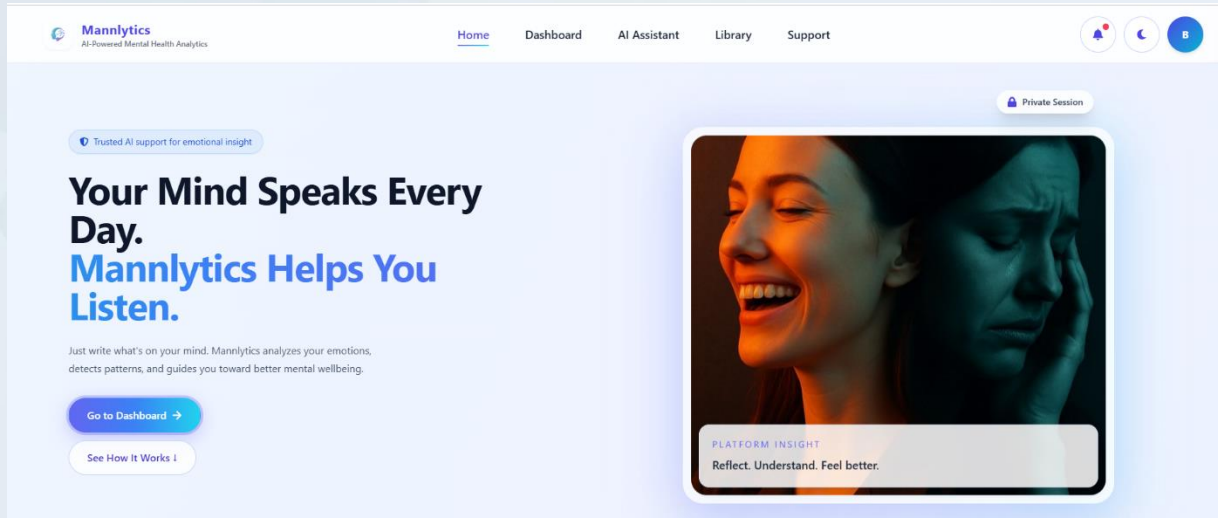
Testing & Results



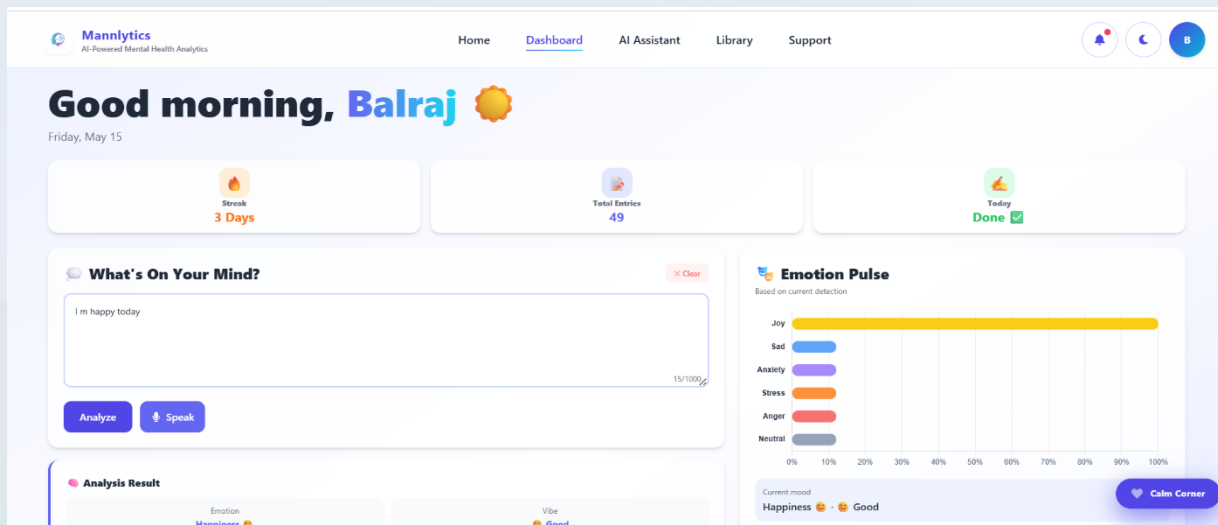
- Evaluated on **52,678 labelled text samples**.
- Compared TF-IDF + Logistic Regression, DistilBERT, and RoBERTa.
- RoBERTa achieved **89.97% accuracy**.
- Predicts seven mental health categories.
- Generates confidence and risk scores.
- Provides real-time predictions with dashboard visualisations.



Demo



Home Page



Dashboard



Detailed Analysis

Friday, May 15 - Confidence: 99%

Your Entry

I'm happy today

AI Insight

Your words suggest a positive and emotionally balanced state.

happy today

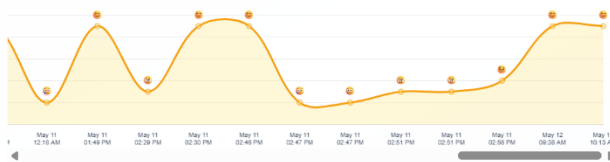
Emotion
Happiness

Predicted Label
Normal

Mental Score
9/10

Wellness Index
85.7
Healthy

Mood Trend



Your emotional pattern looks balanced with some variation.

— swipe to explore —

Mood Highlights



Recommendations

- Keep up your current positive habits and routines.
- This is a great time to set a new goal or help someone around you.
- Write down what is working well so you can revisit it on harder days.

Detailed Analysis

← Back Dark Mode

Export Full Library PDF

MENTAL HEALTH LIBRARY

11 topics - Free to read

What are you trying to understand?

This library covers 11 mental health topics — from everyday emotions to clinical conditions. Read symptoms, lived experiences, coping strategies, and know when to seek help.

Feeling low? Feeling anxious? Stressed out? In crisis?

TOPICS
11

SAVED
0

CATEGORIES
4

Explore

Search topics...

FILTERS

All (11) Condition (8)
Emotion (3) Wellness (1)

Quick Mental Health Tips

Small daily actions that support emotional balance.

Drink water first thing in the morning — dehydration can worsen fatigue, mood, and focus.

Step outside for 10 minutes daily — natural light can support sleep rhythm and emotional.

Reduce screens before bedtime — better sleep often improves mood regulation.

Pause for 3 slow breaths before reacting — it gives your nervous system a chance to settle.

Name your feeling in one sentence each day — awareness usually comes before action.

TIP OF THE DAY

Drink water first thing in the morning — dehydration can worsen fatigue, mood, and focus.

Library

Future Scope

Multi-Language Support – Enable the platform to analyze and respond in multiple languages to reach a broader user base.

Wearable Device Integration – Use data from smartwatches and fitness trackers (e.g., sleep and heart rate patterns) to enhance mental health assessment.

Advanced Predictive Analytics – Incorporate more sophisticated deep learning models and longitudinal analysis for improved prediction accuracy.



Conclusion

- Developed an AI-based platform for mental health detection using NLP.
- Achieved **89.97% accuracy** in classifying seven mental health categories.
- Provides early detection, risk scoring, and personalized recommendations.
- Includes chatbot, voice input, dashboard visualizations, and Calm Corner.
- Supports mental health awareness and early screening, but does not replace professional diagnosis.





THANKYOU

